

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering (profiling, cache-aware design, vectorization, multithreading on Linux), Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

C++, Python, Bash, JavaScript, TypeScript, Java, SQL

Systems & Performance

Linux `perf` profiling, cache-aware design, OpenMP, AVX intrinsics, SIMD, distributed-systems concepts, async services, low-latency pipelines, integration & load testing, Spark / PySpark, Spark UI, Databricks

Backend & Reliability

Node.js, Express, FastAPI, REST APIs, authentication hardening, rate limiting, audit logging, multi-tenant SaaS, VAPT remediation

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Cloudflare Workers (personal project)

Databases & Tools

MongoDB, PostgreSQL, Redis, SQLite, GTest / CMake

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned backend reliability and security across **multi-tenant NHS eTriage systems** (Node.js, MongoDB) serving **8+ UK clients** and **10,000+ monthly submissions** integrated with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Debugged end-to-end production issues across releases and tenant integrations on Linux: monitored production logs, root-caused tenant-specific incidents, and shipped fixes; improved operational stability through authentication hardening, rate limiting, and audit logging.
- Remediated **20+ VAPT findings** and added integration tests to protect sensitive clinical workflows against regressions; enforced secure-by-default patterns across tenant isolation and backend access paths.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.

Projects

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Databricks, SQL, Python

- Optimized a distributed Spark ETL pipeline on the NYC TLC taxi dataset by tuning broadcast joins, caching, partitioning, adaptive Spark settings, and column pruning to reduce shuffle volume and wall-clock runtime.
- Profiled stage-level timings via **Spark UI** to localize shuffle and skew bottlenecks across naive vs. tuned runs and documented a repeatable performance-tuning workflow for distributed data jobs.

Sonare - Offline Real-Time Audio/Video Pipeline

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Engineered the FastAPI backend and Electron/React frontend to run **fully offline** on local hardware; designed **async APIs and queues** for low-latency video and audio stream processing without external network dependencies.
- Integrated MediaPipe hand-tracking, whisper.cpp ASR, and local TTS into an on-device pipeline coordinated through the FastAPI service, end-to-end product designed and shipped under hackathon time pressure.

AgenticSearch - Provenance-First Service with Instrumentation

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Hardened a multi-stage Python/FastAPI service with **150+ automated tests**, provider routing and fallback across upstream backends, pipeline instrumentation, and a reviewer-facing trust UI for review-friendly debuggability and validation.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.